



Accredited with **A** Grade by **NAAC**

12-B Status from UGC

BCA IVth Semester

Computer Communication and Networks

Course Code: BCAC0011

Presented by:

Dr Anuj Mangal

Assistant Professor, Dept. Of CEA

GLA University, Mathura

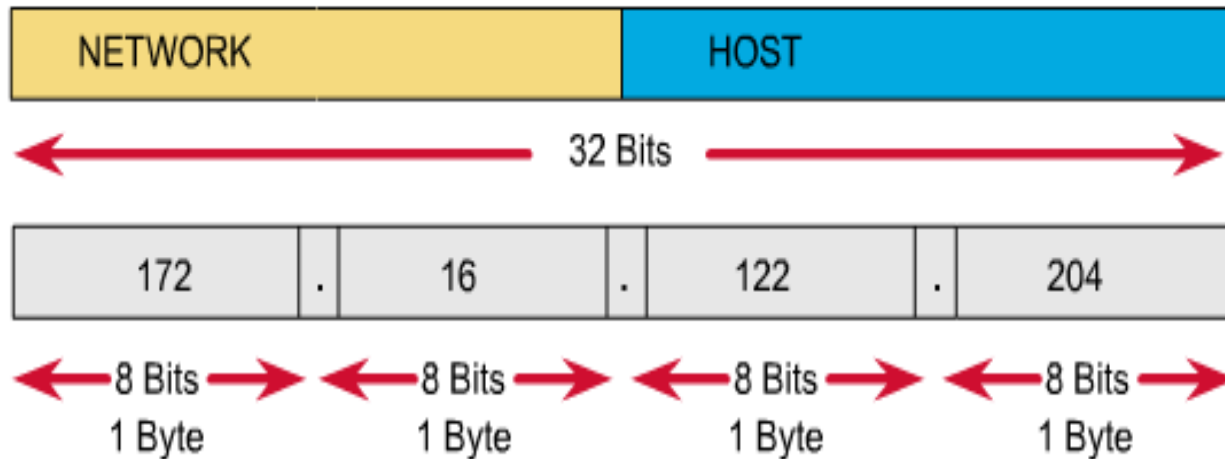
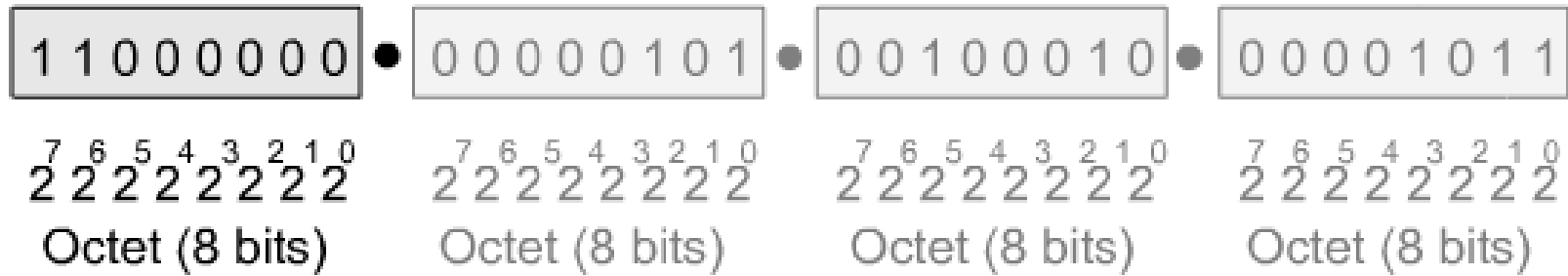
—

Basics

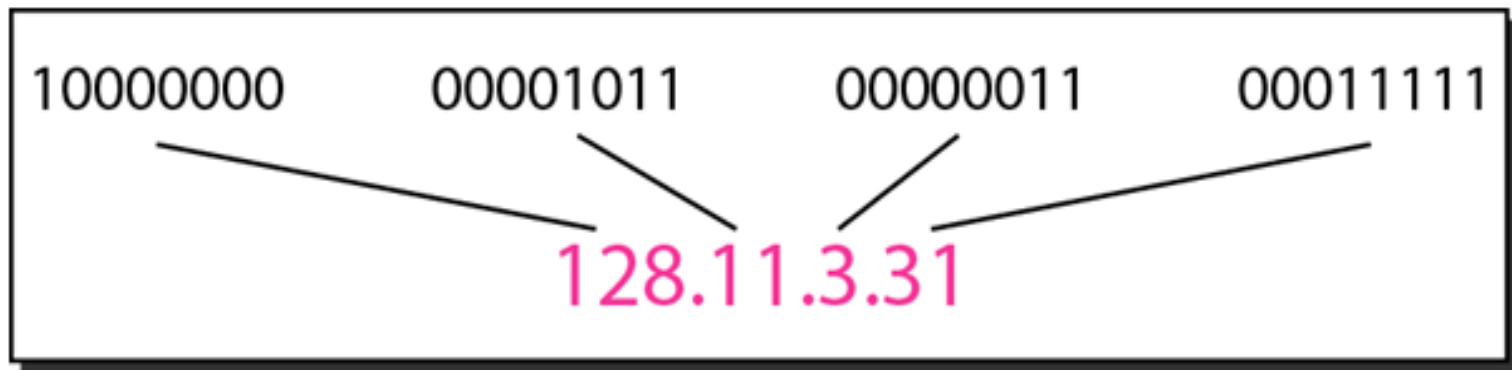
- The IPv4 addresses are unique and universal.
- An IPv4 address is 32 bits long.
- The address space of IPv4 is 2^{32} (4,294,967,296)
- 32 bit address is divided into 4 octet (also known as byte)
- To connect on internet each device requires a unique IP address

IPv4 Addressing

IP Address as a 32-Bit Binary Number (Four Octet)



Binary to DDN conversion



Example 1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

- a. 10000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 11100111 11011011 10001011 01101111
- d. 11111001 10011011 11111011 00001111

Example 1 Solution

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

- a. 10000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 11100111 11011011 10001011 01101111
- d. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation:

- a. 129.11.11.239
- b. 193.131.27.255
- c. 231.219.139.111
- d. 249.155.251.15

Example 2



GLA
UNIVERSITY
MATHURA
Recognised by UGC Under Section 2(f)

Accredited with **A** Grade by **NAAC**

12-B Status from UGC

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

- a. 111.56.45.78
- b. 221.34.7.82
- c. *241.8.56.12*
- d. *75.45.34.78*

Example 2 Solution

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

- a. 111.56.45.78
- b. 221.34.7.82
- c. 241.8.56.12
- d. 75.45.34.78

Solution

We replace each decimal number with its binary equivalent:

- a. 01101111 00111000 00101101 01001110
- b. 11011101 00100010 00000111 01010010
- c. 11110001 00001000 00111000 00001100
- d. 01001011 00101101 00100010 01001110

Example 3

Find the error, if any, in the following IPv4 addresses:

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Example 3

Find the error, if any, in the following IPv4 addresses:

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

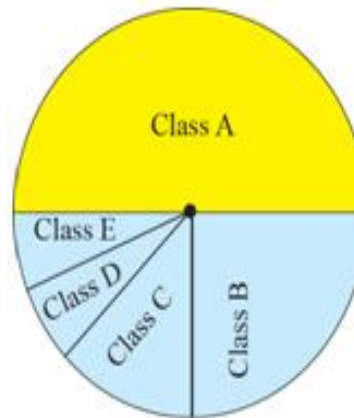
Solution

- a. There should be no leading zeroes (045).
- b. We may not have more than 4 bytes in an IPv4 address.
- c. Each byte should be less than or equal to 255.
- d. A mixture of binary notation and dotted-decimal notation.

IPv4 Addressing Classification

IPv4 addresses can be classified on 2 ways:

- Classful Addressing
- Classless Addressing



Class A:	2^{31}	= 2,147,483,648 addresses, 50%
Class B:	2^{30}	= 1,073,741,824 addresses, 25%
Class C:	2^{29}	= 536,870,912 addresses, 12.5%
Class D:	2^{28}	= 268,435,456 addresses, 6.25%
Class E:	2^{28}	= 268,435,456 addresses, 6.25%

In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.

Classful Addressing

- In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Class	Net ID bits	Host ID bits	Binary	DDN	No of networks	No of host/Nw	Default mask	CIDR
A (2^{31})	8	24	0	0-127	$2^7=128$	2^{24}	255.0.0.0	/8
B (2^{30})	16	16	10	128-191	2^{14}	2^{16}	255.255.0.0	/16
C (2^{29})	24	8	110	192-223	2^{21}	2^8	255.255.0.0	/24
D (2^{28})	NA	NA	1110	224-239	NA	NA	NA	NA
E (2^{28})	NA	NA	1111	240-255	NA	NA	NA	NA

CIDR = classless Interdomain Routing Notation

DDN = Dotted Decimal Notation

Find the class of each address.

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 14.23.120.8
- d. 252.5.15.111

Solution

- a. The first bit is 0. This is a class A address.*
- b. The first 2 bits are 1; the third bit is 0. This is a class C address.*
- c. The first byte is 14; the class is A.*
- d. The first byte is 252; the class is E.*

- **Thank You**
- **Discussion & Doubt session will be in lecture.**

Important terms to keep in mind

1. **IP Address**
2. **IPv4 Binary and DDN notation**
3. **Classful Addressing**